Supplementary Materials

A cell-type-resolved atlas of deep-space stress susceptibility in the dry and germinating Arabidopsis seed
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Contents: Supplementary Methods · Supplementary Tables S1–S6 · Supplementary Figures S1–S5 · Evidence-tier audit. All tables are provided as machine-readable files in the code repository (paths given); key items are summarised inline. Figures are reproduced by `scripts/20`–`scripts/26`.

Supplementary Methods

Seed single-cell reference. Developmental atlas: Gehring lab snRNA-seq, 3/5/7 DAP (GEO GSE295007; 23,374 genes × 54,210 nuclei; level_1 5 tissues → level_2 13 cell types → level_3 85 states + 43 GO module scores). Germination atlas: Liew/Lewsey scRNA-seq, 12/24/48 h (matrix via GEO mirror GSE182331; 13,501 genes × 12,798 cells; 15 clusters named from the source paper, clusters 9/14 in-situ validated). Marker panels: Wilcoxon `rank_genes_groups`, top-50 genes/group, ≥20 cells → 122 panels (GMT in `panels/panel_library.gmt`). The **dry/0 h pole** is anchored by a bulk-tier `dry_seed` panel (top-50 genes up in mature dry seed, 21 vs 15 DAP, GSE76015, mean of 3 WT ecotypes) → 123 panels total.

Stressor signatures (27 contrasts, 10 families). Genome-wide differential expression / treatment-vs- control signatures harmonised to AGI/TAIR (Entrez, Affymetrix ATH1/GPL198 and Agilent GPL9020 maps in `data/raw/`). Sources in Table S1/S2. Null-magnetic-field (NMF) handled by **expression-localization** (panel too small for GSEA; ≤7-gene marker overlap): NMF-up/down gene specificity (z) vs 1,000 random sets; two panels (2022 oxidative; 2021 Sci Rep S7 DEGs).

Projection (DSRS / GSAD). Each signature ranked by log2FC and projected onto the panels by pre-ranked GSEA (`gseapy`; 200 permutations; seed 42) → NES + FDR per panel. DSRS scores query↔reference fingerprint similarity (rank correlation). GSAD returns the per cell-type/tissue/stage susceptibility profile.

Bridge (shared latent). Germinating-seed score space (15 axes). Adult stress placed by decoder NES; late-seed-dev by ssGSEA of Gehring pseudobulk. **Within-source scaling** removes a score-type artefact (PC1↔source |r| 0.38→0.00). Embryo-lineage variant recovers textbook developmental lineages (positive control).

Atlas / convergence. A cell type/stage is "susceptible" to a family if any contrast in that family has |NES| ≥ 1.5 & FDR < 0.25 (or NMF localization |z| ≥ 2). Convergence = number of the 10 families meeting the threshold (family-level voting prevents data-rich families from dominating).

Software / reproducibility. `deepspace` Python package (DSRS + GSAD + CLI); numbered scripts 01–26; deterministic seeds; `REPRODUCE.md`. MIT license.

Supplementary Tables (machine-readable in repo)

- **Table S1 — Data inventory** (all datasets, accessions, platform, role, evidence tier, caveats): `data/data_inventory.csv`.
- **Table S2 — Stressor contrast manifest** (27 contrasts: family, dataset, dose/time, genotype): `results/tables/contrast_classes.csv` + `radiation_contrasts_manifest.csv`.
- **Table S3 — Decoder NES/FDR matrix** (123 panels × 27 contrasts): `results/tables/decoder_nes_matrix_v7.csv`, `decoder_fdr_matrix_v7.csv`.
- **Table S4 — Atlas convergence** (cell-type and tissue/stage): `results/tables/deepspace_atlas_convergence.csv`, `deepspace_atlas_tissue_stage_convergence.csv`.
- **Table S5 — NMF localization** (2022 + 2021 panels): `results/tables/nmf_localization.csv`, `nmf2021_localization.csv`.
- **Table S6 — Panel library** (123 seed cell-type/state panels): `panels/panel_library.csv` / `panel_library.gmt`.

Evidence-tier audit (key claims)

Claim	Tier	Basis	Falsification
Root/radicle tip = top multi-stressor hotspot (columella 9/10, radicle meristem 6/10)	D×A→H	genome-wide DGE projected onto validated panels	per-stressor knock-down / spaceflight germination; predict root-tip-program shift
Early germination (12 h) = most vulnerable stage	D→H	family convergence over stage panels	tissue-matched stress panel; predict 12 h > 24/48 h
NMF-up genes → radicle apical meristem (z +7.96)	L×A→H	expression localization (2022 oxidative panel)	NNMF germination + cry1cry2; radicle-meristem markers
NMF radicle signal is oxidative-panel-specific	L×A	2021 DEG panel does not reproduce it (r=0.35)	full genome-wide NNMF arrays (pending)
Embryo lineage recovered (hypophysis→radicle meristem etc.)	A	positive control, transcriptome-only	lineage tracing / marker persistence
Dry-seed program induced by desiccation, suppressed by ethylene	D×A	decoder NES on dry_seed panel	ABA/ethylene dose; dormancy markers

All atlas/decoder outputs are **predicted concordance (signature transfer), not proven causation.**

Supplementary Figures

(embedded below; high-res PNG+SVG in `results/figures/`)

- **Fig S1** — Combined 27-contrast perturbation model (10 stressor families) → seed programs.
- **Fig S2** — Shared-latent bridge (within-source scaled): heatmap + PCA embedding.
- **Fig S3** — Atlas at tissue + germination-stage level (incl. dry-seed anchor).
- **Fig S4** — NMF-responsive gene localization (2022 oxidative panel).
- **Fig S5** — NMF localization cross-check: 2022 oxidative panel vs 2021 Sci Rep NNMF DEGs.

Supplementary Figures

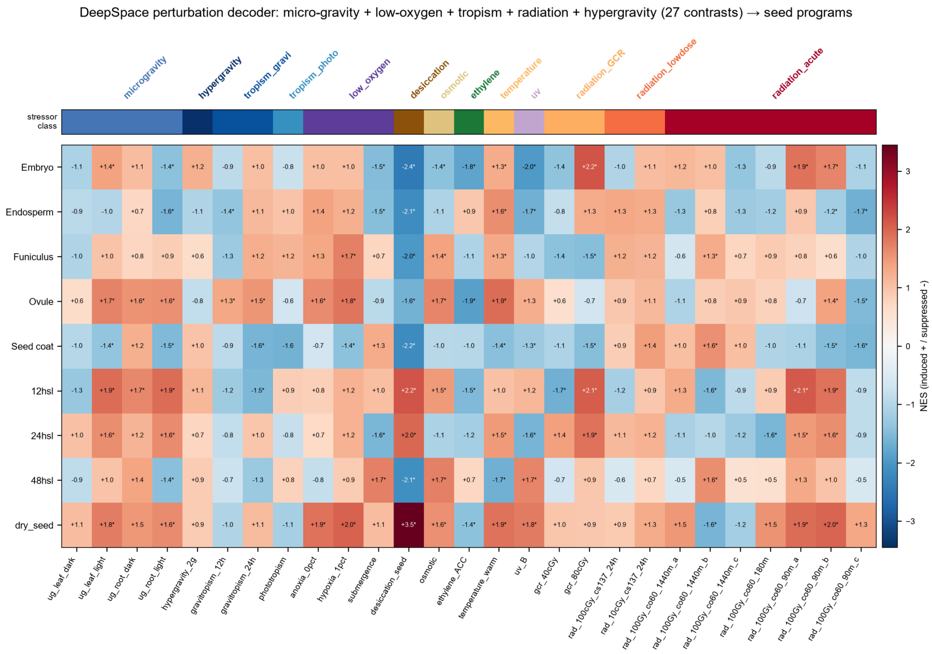

Fig S1. Combined 27-contrast perturbation model (10 families) -> seed programs.

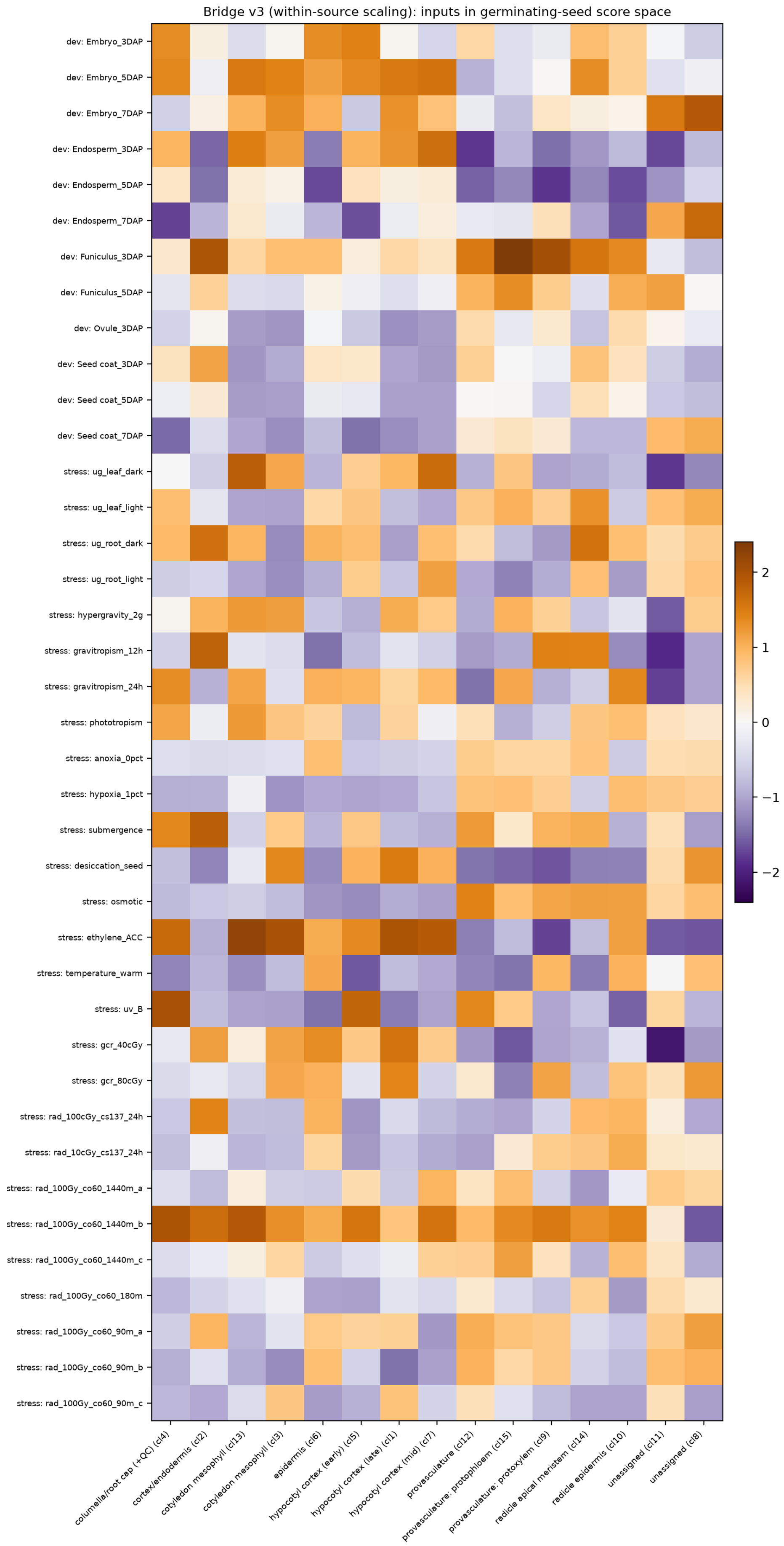

Fig S2a. Shared-latent bridge (within-source scaled).

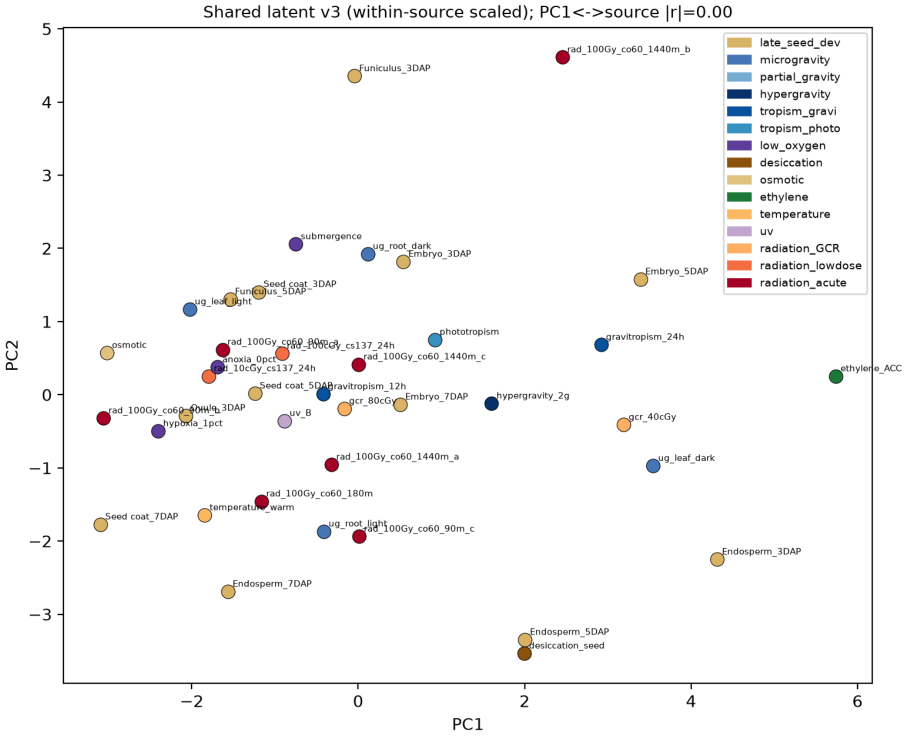

Fig S2b. Shared-latent bridge embedding (PCA).

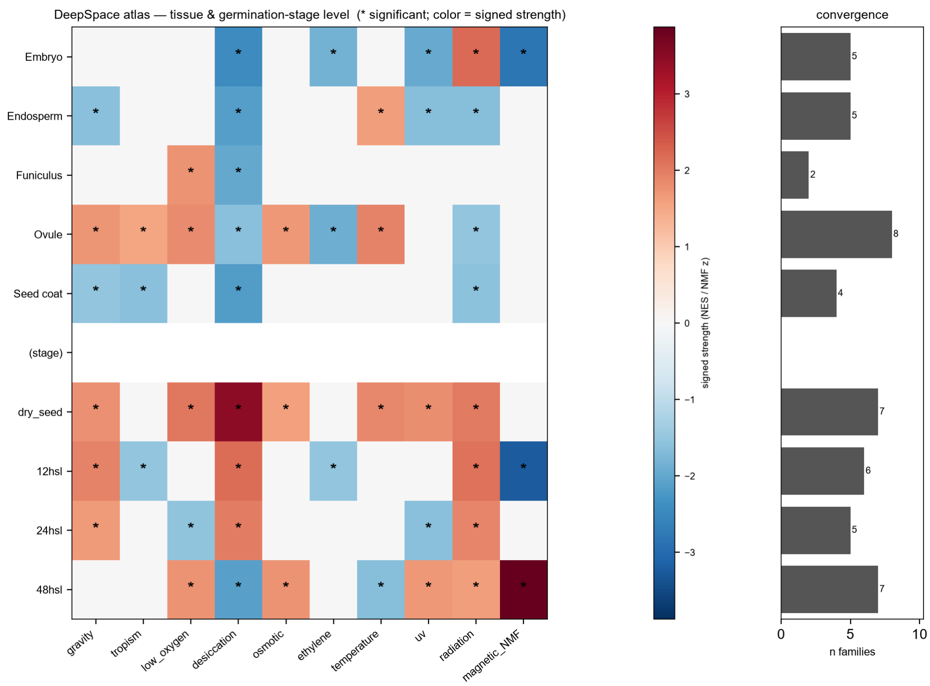

Fig S3. Atlas at tissue + germination-stage level (incl. dry-seed anchor).

Fig S4. NMF-responsive gene localization (2022 oxidative panel).

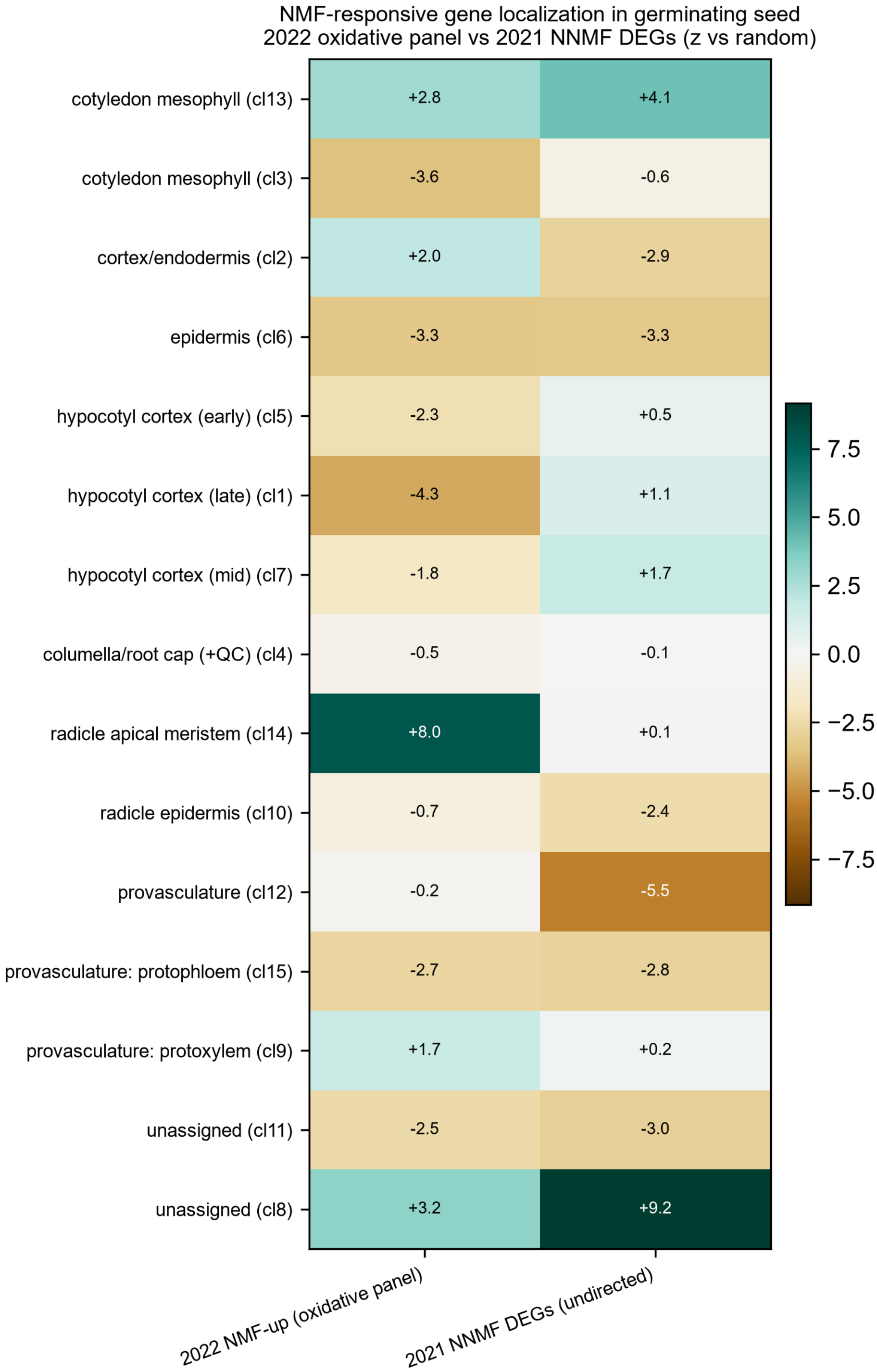

Fig S5. NMF localization: 2022 oxidative panel vs 2021 Sci Rep NNMF DEGs.